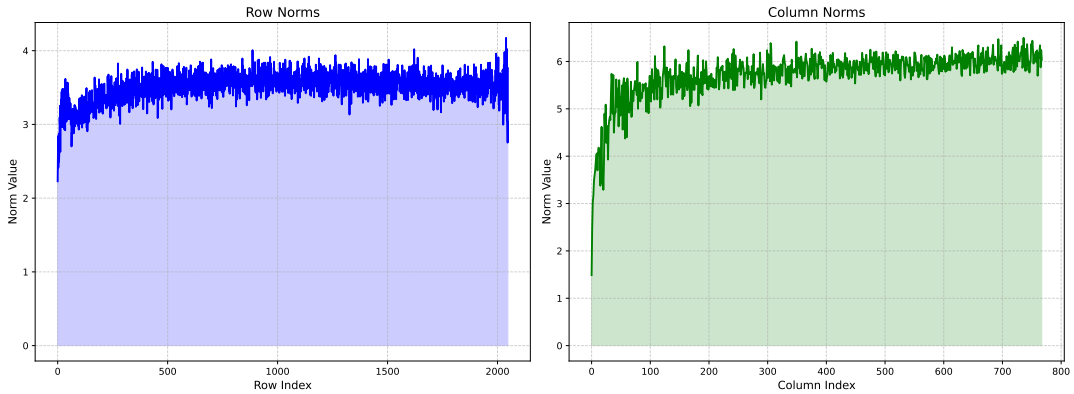
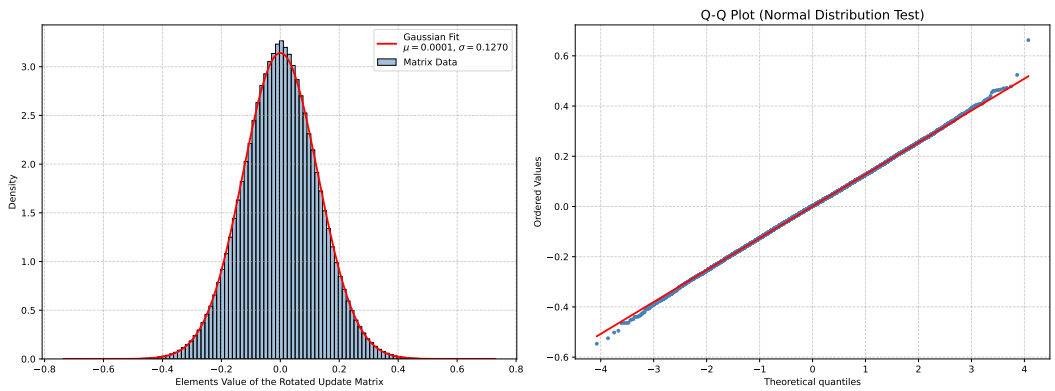
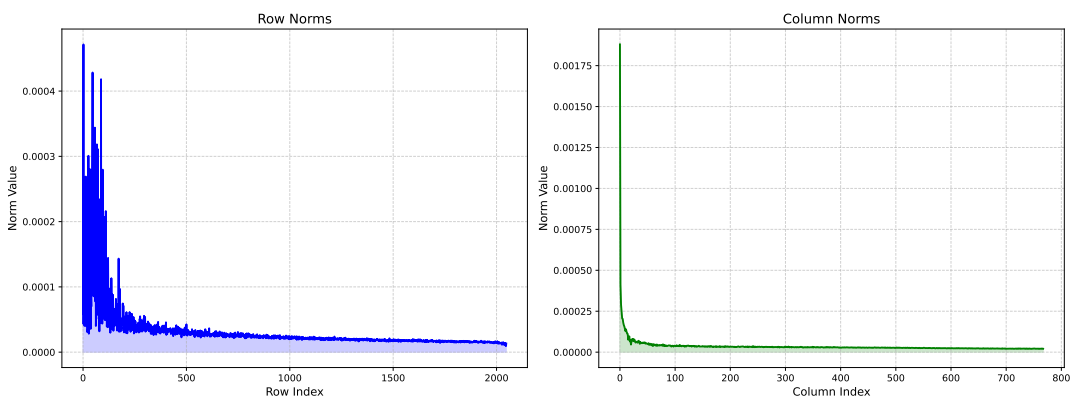


((a)) Row and Column norms of SOAP update $\frac{M_{rot}}{\sqrt{V}}$ matrix in the rotated space.



((b)) Left: Distribution of the update matrix elements in the rotated space, which closely matches an independent Gaussian fit. Right: The Q-Q plot for the distribution shown on the left. The adherence to the straight line validates the Gaussianity of the elements.



((c)) Row and Column norms of SOAP momentum (M_{rot}) matrix in the rotated space.

Figure R1: Visualization of SOAP update statistics. The figure illustrates the underlying metrics of a gate_proj block of a 130M LLaMA model at training step 5000, capturing the mid-to-late stage dynamics.